LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	19
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Pneumatic system basic design requirements, brief description, advantages and disadvantages of a typical pneumatic system.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Pneumatic system basic design requirements, brief description, advantages and disadvantages of a typical pneumatic system.
5. Specific Objectives	17.1.4.1 Trainee will be able to describe the construction and functioning of a typical pneumatic system and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB diagrams; Instructor supervision required for safety regarding high-pressure system concepts.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary working fluid in a pneumatic system?
(b) Name one major advantage of pneumatic systems over hydraulic systems.
(c) What is the basic design requirement for air supply in a pneumatic system?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic design requirements for pneumatic systems • Brief description of typical pneumatic system layout • Functioning of air compressor and storage components • Role of pressure regulators and filters in system • Operation of pneumatic actuators and valves	L & Ds	PPT, IB	15 min
(b)	• Advantages of pneumatic systems in aircraft • Disadvantages and limitations of pneumatic systems • Comparison with hydraulic and electrical systems • Safety considerations for pneumatic operation • Maintenance implications of pneumatic components	L & Ds	PPT, PDF	20 min
(c)	• Summary of typical pneumatic system construction • Key functional components and their roles • System limitations and operational constraints • Future trends in pneumatic system design • Integration with other aircraft systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) List three advantages of using compressed air in aircraft systems.
(b) Explain why moisture removal is a critical design requirement in pneumatic systems.
(c) Describe the primary function of a pressure regulator in a typical pneumatic system.

12. Home Assignment and Follow Up	Prepare a brief note on how the knowledge of pneumatic system design requirements will assist in understanding the specific purpose and operation of typical pneumatic system components covered in LP 20.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Description purpose typical pneumatic system components	Refer to the master syllabus reference list	LP 20

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min